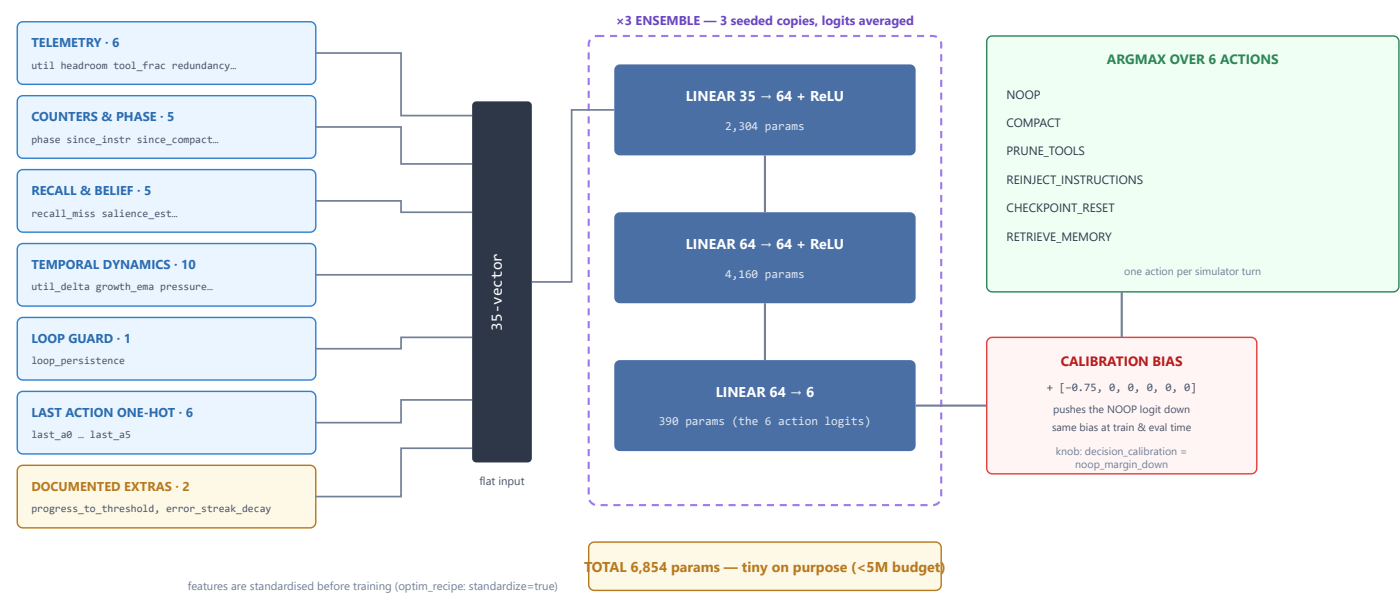


What we're dissecting: **exp_003** — the selected best (judge 82, sim 82.28). The deploy bundle ships **exp_010** (the hidden-exam winner), which uses the **same 35 features** and the **same 35→64→64→6 architecture**; its card is the source of the verbatim feature list below.

THE PIPELINE — FROM TELEMETRY TO ACTION

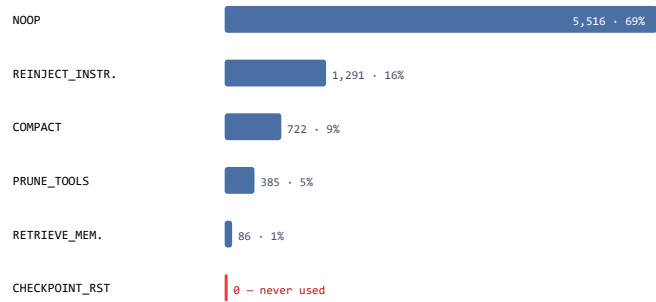


ALL 35 FEATURES — VERBATIM FROM THE DEPLOY CARD

TELEMETRY (6)	util · headroom · tool_frac · redundancy · stale · confidence
COUNTERS & PHASE (5)	phase · since_instr · since_compact · error_streak · last_error
RECALL & BELIEF (5)	recall_miss · topic_shift · salience_est · integrity_est · recall_intensity
TEMPORAL DYNAMICS (10)	util_delta · growth_ema · steps_to_overflow · pressure · streak_pressure
	tool_rate · shift_rate · cliff_margin · required_rate · recent_success
LOOP GUARD (1)	loop_persistence
LAST ACTION 1-HOT (6)	last_a0 · last_a1 · last_a2 · last_a3 · last_a4 · last_a5
DOCUMENTED EXTRAS (2)	progress_to_threshold · error_streak_decay

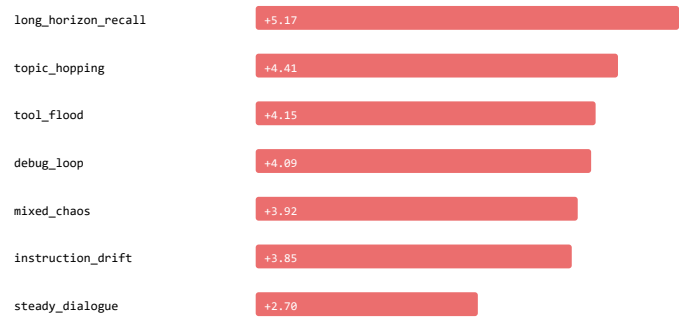
Groups are for readability — the model sees one flat 35-vector. Design notes: bounded history windows (history[-8:], [-6:]), every value passed through `_safe()` for finite floats, batch-size-1 safe forward.

ACTION MIX — 8,000 DECISIONS



Judge: RETRIEVE_MEMORY fires almost only in long_horizon_recall turns; NOOP share stays 69% even with the -0.75 bias

GAP TO THE PERFECT-INFO CEILING, PER FAMILY



return points below clairvoyant · smaller = closer to perfect · worst family is memory, easiest is steady chat

HOW IT WAS TRAINED — THE RECIPE

piece	setting
data	behaviour cloning + DAgger: 84 self-play episodes, relabelled, retrained
teacher	v2 — two-step lookahead (labels + DAgger relabelling)
target	soft distillation, softmax(q / 0.5)
optimizer	adamw · lr 5e-4 · 30 epochs · batch 256 · weight_decay 0.01 · grad_clip 1.0
ensemble	3 seeded copies, logits averaged
features	35, standardised

success 98%

overflow 0.0

mean return +33.51

oracle agreement 51.9%

WHAT THE JUDGE FOUND IN THE LOG

No overfit: train_acc 59.0% vs val_acc 59.6% — the gap is negative.

DAgger converged: round-1 return on its own self-play episodes was +32.77, vs +33.51 deployed.

Weakest families: debug_loop 0.89 and topic_hopping 0.96 success — everything else $\geq \sim 0.97$.

Beats its teacher: +33.51 vs teacher_v2's +32.14 and the one-step oracle's +31.73 in aggregate.

Oracle agreement only 51.9% — it wins by context management, not by mimicking the oracle; dips to 0.25–0.35 on some tool_flood / instruction_drift turns.

Judge's verdict: base score = round(sim_score) = 82, no deduction flags (no NOTE lines, overflow 0.00 < 0.02 threshold).

READ IT THIS WAY

A tiny 6.9k-param MLP on 35 hand-documented features, distilled from a two-step lookahead teacher, corrected by 84 rounds of its own mistakes (DAgger), with the "do nothing" option deliberately damped — and it out-scores the teacher it was distilled from.

THE TWO CHAMPIONS — DON'T MIX THEM UP

exp_003 — SELECTED BEST

Crowned by the spec rule: highest judge score (82), tie broken by sim (82.28). Eval return **+33.51**. Lives at workspace/best_model.py. This page's metrics and judge evidence describe this one.

exp_010 — CONFIRMED BEST

Won the hidden exam: 200 fresh scenarios, seeds 95000+, never used for selection → sim **81.37**, return **+32.93**. Its card lists the same 35 features and ensemble×3. Lives at workspace/best_model_confirmed.py.

WHAT SHIPS — deploy/

The bundle deploys **exp_010** (the evidence winner): intervention_policy.pt + runtime + policy_card.json. Parity check on the card: **all 50 replay turns identical** between simulator and runtime.